Automated Podcast Transcription & Topic Segmentation

Transforming Unstructured Audio into Searchable, Structured Knowledge

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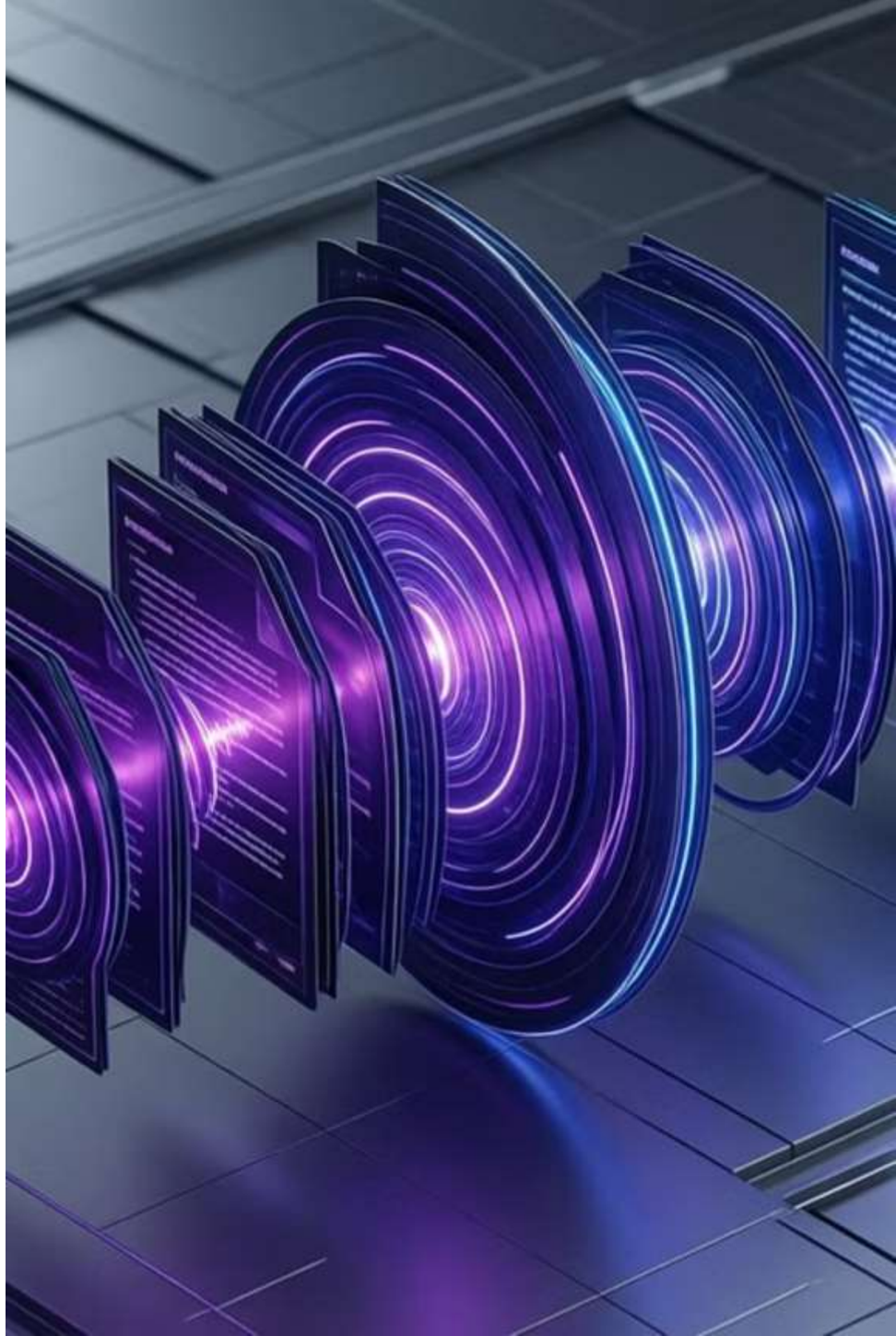


The Problem Statement

- **The "Unstructured Data" Crisis:** Podcasts and lectures generate massive amounts of audio that cannot be searched or skimmed.
- **The Efficiency Gap:** Finding a specific 5-minute discussion within a 2-hour episode is tedious and manual.
- **Current Limitations:**
  - Manual transcription is slow and expensive.
  - Existing tools provide a "wall of text" without logical chapter divisions.

Project Objectives

1 Precision Transcription:

Achieve near-human accuracy in speech-to-text using **OpenAI Whisper**.

2 Intelligent Segmentation:

Use **NLTK TextTiling** to detect natural topic boundaries based on context.

3 Smart Summarization:

Deploy **BART** to generate concise abstractive summaries for each topic.

4 Keyword Discovery:

Extract semantic keywords using **KeyBERT** for quick scanning.

System Architecture

Frontend Layer:

Streamlit dashboard for user interaction and visualization.

Processing Layer:

Python pipeline managing audio normalization and model inference.

AI Core:

- **Whisper:** ASR (Automatic Speech Recognition).
- **BART:** Abstractive Summarization.
- **KeyBERT:** Keyword Extraction.

Data Layer:

JSON/CSV storage for structured results.

Project Workflow

1

Ingestion:

User uploads MP3/WAV files.

2

Preprocessing:

Silero VAD removes silence; **Librosa** normalizes audio to 16kHz.

3

Transcription:

Whisper converts audio to timestamped text.

4

Segmentation:

TextTiling analyzes lexical cohesion to find topic shifts.

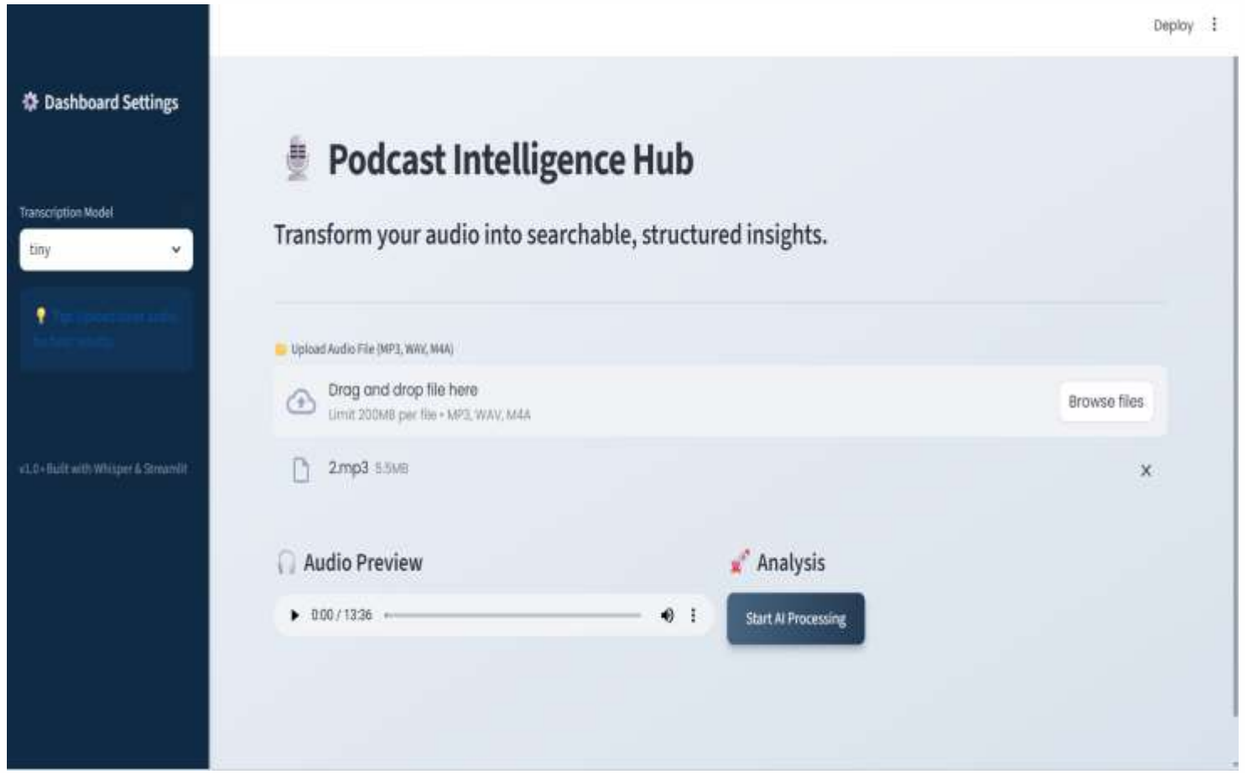
5

Analysis:

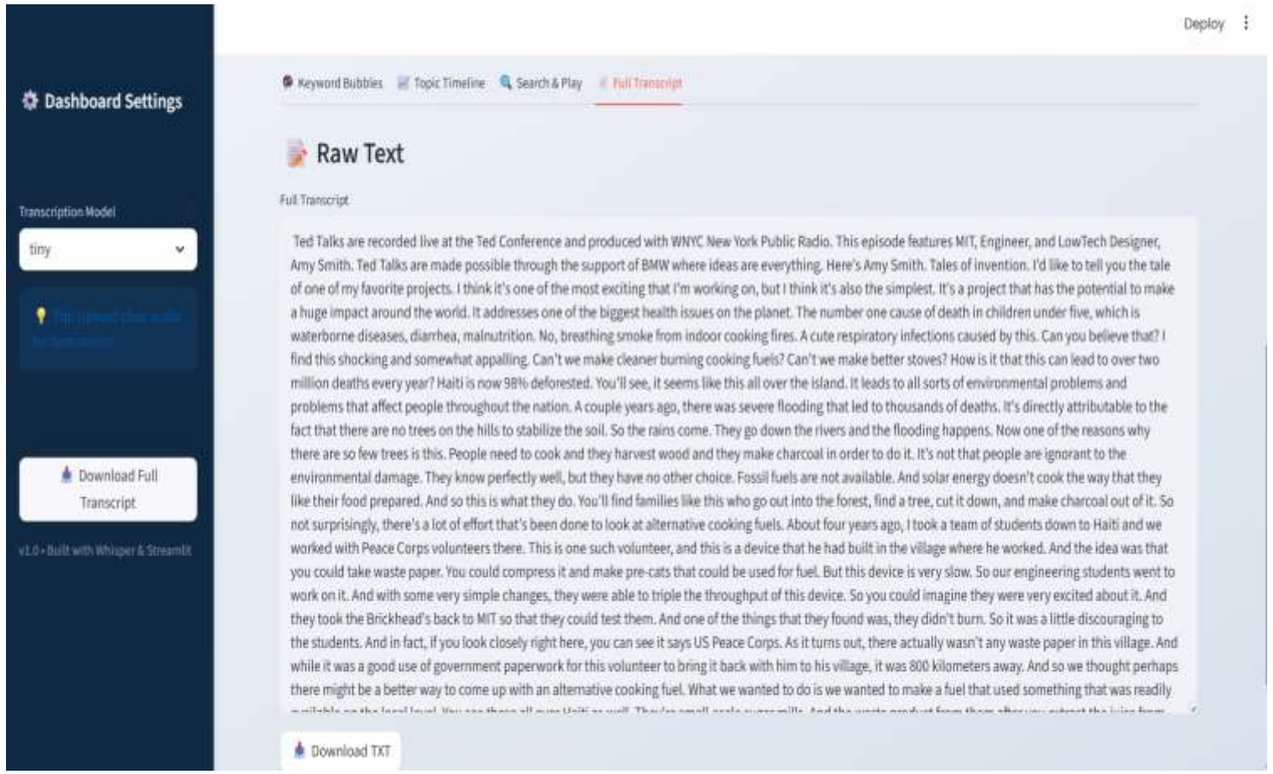
BART summarizes the segment; **KeyBERT** extracts tags.

User Interface

Dashboard

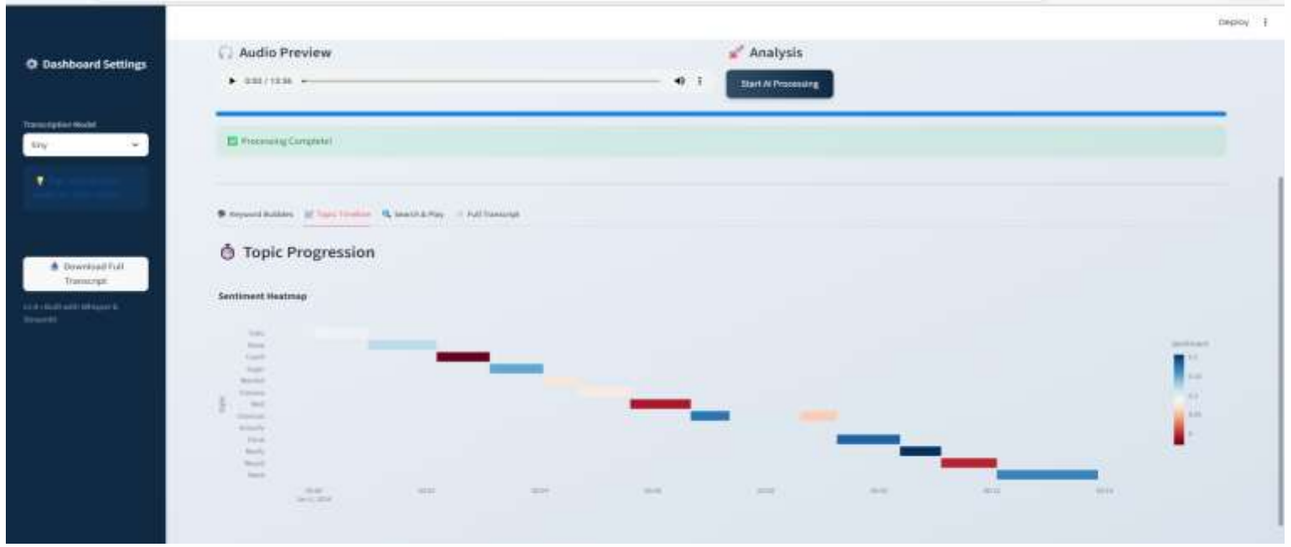


Full Transcript

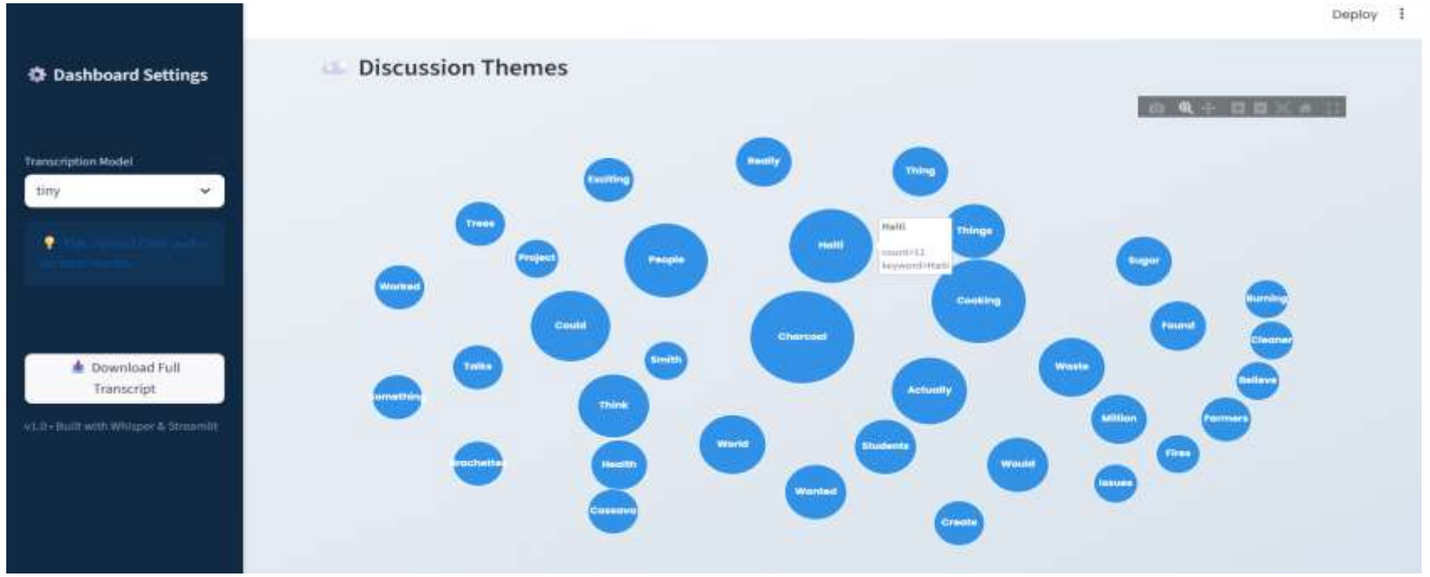


Segments & Search Interface

Sentiment Timeline



Keyword Bubbles



Tech Stack

Core Language:

- Python 3.9+

Audio Processing:

- **Silero VAD:** Enterprise-grade silence removal.
- **Librosa:** Signal processing & normalization.
- **FFmpeg:** Format conversion.

AI Models:

- **OpenAI Whisper:** Transformer-based ASR.
- **BART (HuggingFace):** Abstractive Summarization.
- **KeyBERT:** Semantic Keyword Extraction.
- **NLTK:** TextTiling Algorithm.

Key Features

Automated Timestamping:

Every segment is mapped to precise audio start/end times.

Abstractive Summaries:

Users get a 3-sentence overview of each topic (via BART).

Semantic Tagging:

KeyBERT extracts themes based on meaning, not just word count.

Interactive Navigation:

Click a topic title to jump to that part of the audio.

Performance & Limitations

Performance:

- **Silero VAD** reduced processing time by ~15% by removing silence.
- **Whisper Base** offered the best balance of speed vs. accuracy.

Limitations:

- **Memory Usage:** BART and Whisper models are RAM-intensive.
- **Speaker Diarization:** The system currently does not distinguish between Speaker A and Speaker B.

Conclusion

- **Success:** Built a fully functional "Audio Intelligence" pipeline.
- **Impact:** Reduces content consumption time by transforming linear audio into structured data.
- **Scalability:** The modular design allows easy swapping of models (e.g., upgrading BART to GPT-4).
- **Final Word:** A practical implementation of modern AI to solve a real-world unstructured data problem.

Thank You
Q/A